Sim-and-Human Co-training for Data-Efficient and Generalizable Robotic Manipulation

Paper-ID 12

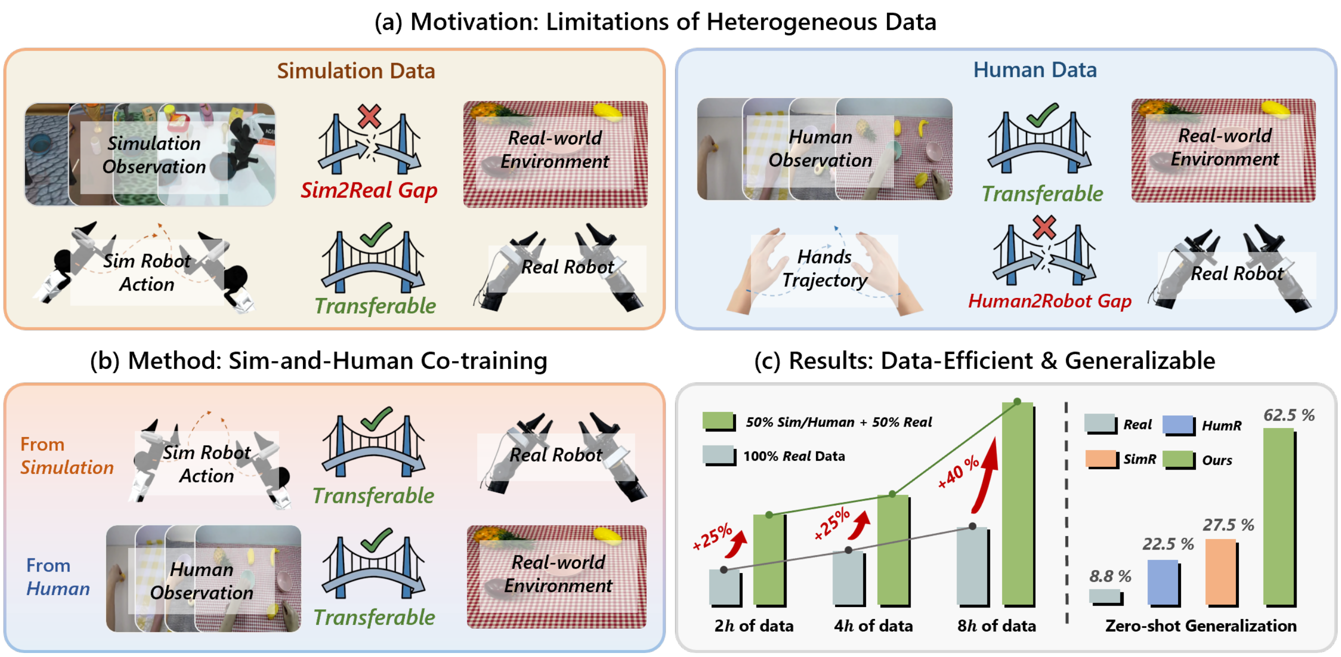

Fig. 1: We propose **SimHum** for Data-Efficient and Generalizable Robotic Manipulation. **(a)** Simulation data and human data each have their own advantages and limitations. **(b)** we present SimHum to integrates the respective strengths of both data sources to establish a generalizable prior. **(c)** Consequently, our approach demonstrates exceptional data efficiency, as detailed in Figure 7(a), and strong generalization capabilities, as reported in Table I.

***Abstract*—Synthetic simulation data and real-world human demonstrations provide scalable alternatives to circumvent the prohibitive costs of robot data collection. However, prior works mostly utilize these sources separately. Consequently, they are either limited by the *human-to-robot embodiment gap* when leveraging human data, or hindered by the *sim-to-real visual gap* when utilizing simulation. In this work, we identify a natural yet widely neglected synergy between these sources: simulation offers the robot action that human data lacks, while human data provides the real-world observation that simulation struggles to render. Motivated by this insight, we present SimHum, a co-training framework to simultaneously extract kinematic prior from simulated robot actions and visual prior from real-world human observations. Based on the two complementary priors, we achieves data-efficient and generalizable robotic manipulation in real-world tasks. Empirical results demonstrate the data efficiency and generalizability of SimHum, showing a 40% success rate improvement over the real-only baseline under an 8-hour data collection budget and a 7.1× increase in unseen scenarios on four real-world tasks.**

I. Introduction

A long-standing challenge in robot learning is enabling policies to generalize across complex manipulation tasks in unconstrained real-world settings. While recent foundation models have demonstrated the potential of large-scale pre-training, their success heavily relies on massive datasets. However, unlike the abundant text data available for LLMs [1, 2, 3, 4], collecting high-quality robotic interaction data in the physical world is labor-intensive, unscalable, and prohibitively expensive. This data scarcity fundamentally bottlenecks the development of robust, generalizable robot policies.

To circumvent the high cost of real-world data collection, recent research has pivoted towards scalable alternatives like physics-based simulation [5, 6, 7, 8] and human video demonstrations [9, 10, 11, 12, 11]. However, as illustrated in Figure 1(a), existing methods mostly leverage these sources in isolation, inevitably inheriting their respective intrinsic discrepancies: the *sim-to-real visual gap* and the *human-to-robot embodiment gap*. However, we observe that these sources are intrinsically complementary, where the strengths of one mitigate the limitations of the other. Specifically, simulation data suffers from unavoidable **real-world visual discrepancies** but offers **robot-valid actions**. Conversely, human data provides **real-world visual observations** but exhibits a **robot-incompatible morphology**. This complementary relationship leading us to ask:

> *Can we directly **co-train** on these sources to leverage their complementary strengths, thereby achieving **data-efficient real-world generalization**?*

To answer this, we introduce **SimHum**, a unified framework designed to simultaneously extract robot-valid kinematic prior from **Sim**ulation and photorealistic visual prior from **Hum**an data, as shown in Figure 1(b). To this end, we first design an alignment data collection pipeline that facilitates the unified transfer of these priors to the real-world robot policy. Next, we present a heterogeneous pre-training stage that utilizes source-dependent modules to explicitly extract the distinct strengths of each data source. Finally, we perform real-world fine-tuning via reality-grounded module grafting, which selectively recombines these pre-trained components. By integrating these procedures, we establish a unified and transferable prior, harnessing it to enable efficient and generalizable real-world robotic manipulation.

We empirically evaluate SimHum in real-world environments with three baselines: Real-only (no pre-training), SimR (simulation pre-training), and HumR (human pre-training), all fine-tuned on the same real-robot dataset. As shown in Figure 1(c), SimHum not only ensures exceptional real-world data efficiency but also advanced zero-shot generalizability. Specifically, under a fixed 8-hour budget, SimHum (4h human/sim + 4h real) outperforms the Real-only baseline (8h real) by $\mathbf{40}\%$. Furthermore, in rigorous Out-of-Distribution (OOD) settings, it achieves an average success rate of $\mathbf{62.5}\%$, surpassing the Real-only baseline by $\mathbf{7.1}\times$ and the SimR and HumR baselines by $\mathbf{2.3}\times$ and $\mathbf{2.8}\times$, respectively. To the best of our knowledge, this is the first work to pre-train a unified policy exclusively on simulation and human data, establishing a scalable paradigm for data-efficient and generalizable robot manipulation.

Our main contributions can be summarized as follows:

1) We propose Sim-and-Human Co-training (SimHum), a unified framework integrating a specialized data pipeline and two-stage training for robot manipulation.
2) Through extensive ablations, We identify the complementary roles of simulation and human data and establish a balanced co-training recipe for efficiently scaling heterogeneous datasets.
3) Real-robot evaluations demonstrate that SimHum achieves data-efficient zero-shot generalization, substantially outperforming robot-only and single-source baseline.

II. Related Works

A. Imitation Learning for Robot Manipulation

Imitation Learning (IL), particularly in the form of Behavior Cloning (BC), has emerged as a dominant paradigm for acquiring robust visuomotor policies directly from expert demonstrations [13, 14, 15]. By mapping sensory observations to control actions, IL circumvents the complexities of manual controller design and reward engineering required by reinforcement learning. Recent advancements have demonstrated the remarkable scalability of this paradigm, driven largely by the evolution of policy architectures [16, 17, 18, 19, 20, 21, 22, 23, 24] and data collection systems [25, 26, 27, 28, 29]. However, the prohibitive cost of acquiring large-scale real-world data remains a fundamental bottleneck that limits the generalization of imitation learning policies. To overcome this, we propose a co-training framework that harmonizes the visual and kinematic priors from scalable human and simulation data to achieve data-efficient generalization.

B. Learning from Scalable Heterogeneous Sources

Leveraging scalable data sources has become a standard practice to mitigate the scarcity of real-robot data. Physics-based simulation [5, 6, 7] offers perfect kinematic labels but suffers from the reality gap, typically addressed via Domain Randomization [30, 31] or mixing with real data [32, 33]. In parallel, human video datasets [34] provide diverse real-world visual semantics but face the embodiment gap. While visual representation learning [35, 36] and cross-embodiment imitation [12, 10, 11] attempt to bridge this gap, they often necessitate real-world demonstrations for kinematic alignment. In this work, we introduce a novel co-training recipe that synergistically combines these sources: we leverage simulation to provide the kinematic grounding often missing in human-to-robot transfer, and human data to bridge the visual reality gap inherent in simulation, thereby enabling data-efficient generalization.

C. Dataset Composition in Robot Learning

Effective dataset composition is essential when leveraging heterogeneous sources, such as varying embodiments, simulation, and human demonstrations. While scaling efforts like Open X-Embodiment [37] show promise, naive data aggregation often induces negative transfer due to distribution mismatches. Prior works have analyzed data scaling laws regarding quality and diversity [38, 39, 40, 41, 42, 43] but focus almost exclusively on real-robot mixtures. This leaves the optimal mixing strategies for heterogeneous data largely unaddressed. This work fills this gap by systematically deriving a recipe for composing human and simulation data, aiming to learn a robust prior for data-efficient real-world generalization.

III. Problem Statement and Preliminaries

A. Formalizing Sim-to-Real and Human-to-Robot Gaps

To circumvent the bottleneck of real-world data ($\mathcal{D}_{real}$) collection, we leverage two scalable data sources: simulation ($\mathcal{D}_{sim}$) and human data ($\mathcal{D}_{hum}$). Formally, we denote a dataset from any source domain as $\mathcal{D} = \{(o_t, a_t)\}_{t=1}^{T}$. Here, $a_t \in \mathcal{A}$ denotes the control action, and o_t is drawn from a visual observation distribution $\mathcal{V}(o_t|s_t)$ conditioned on the latent environment state s_t. While these sources provide abundant supervision, they introduce distinct distributional shifts relative to $\mathcal{D}_{real}$:

The *Sim-to-Real* Visual Gap. Simulation provides perfect kinematic alignment with real robot($\mathcal{A}_{sim} \equiv \mathcal{A}_{real}$), but the observation distribution differs significantly with real-world due to the rendering discrepancy and environmental layout mismatches:

$$\mathcal{V}_{sim}(o_t|s_t) \neq \mathcal{V}_{real}(o_t|s_t), \quad \text{s.t. } \mathcal{A}_{sim} \equiv \mathcal{A}_{real} \tag{1}$$

This implies that while the action labels are valid, the visual features suffer from a severe distributional shift.
The *Human-to-Robot* Embodiment Gap. Conversely, as human videos are collected directly within the target deployment environment, they provide real-world visual semantics ($\mathcal{V}_{hum} \approx \mathcal{V}_{real}$). However, the morphological mismatch between the human hand and robot gripper introduces a fundamental embodiment gap:

$$\mathcal{A}_{hum} \neq \mathcal{A}_{real}, \quad \text{s.t. } \mathcal{V}_{hum}(o_t|s_t) \approx \mathcal{V}_{real}(o_t|s_t) \tag{2}$$

Our objective is to bridge these complementary gaps by fusing the kinematic validity of $\mathcal{D}_{sim}$ with the visual fidelity of $\mathcal{D}_{hum}$.

B. Environmental Factorization for Generalization

To systematically investigate the impact of data composition on real-world generalization, we decompose the environmental variation space into explicit factors. We assume the task variation stems from a joint distribution over a factor space:

$$\mathcal{F} = \langle F_{obj}, F_{dist}, F_{light}, F_{bg}, F_{init} \rangle \tag{3}$$

Specifically, we identify five critical axes of variation:

- **Target Object ($\mathcal{F}_{obj}$):** The specific object instance required to complete the task, characterized by its geometry and semantics.
- **Visual Distractors ($\mathcal{F}_{dist}$):** Task-irrelevant objects that introduce clutter and occlusion.
- **Illumination ($\mathcal{F}_{light}$):** Variations in lighting spectrum and intensity.
- **Background ($\mathcal{F}_{bg}$):** Workspace surface attributes, including texture and color.
- **Initial Pose ($\mathcal{F}_{init}$):** The distribution of the target object's initial 6-DoF pose.

Based on this factorization, we explicitly construct the *Zero-Shot evaluation setting* in Section V-B by combining factor instances that are strictly held-out from the training set. Furthermore, this formulation allows us to conduct a factor-wise ablation in Section VI-B to systematically investigate the impact of OOD shifts in each factor on policy performance.

IV. Method

In this section, we present the **SimHum**: **Sim**-and-**Hum**an co-training to capture the robotic kinematics knowledge inherent in $\mathcal{D}_{sim}$ with the real-world visual diversity preserved in $\mathcal{D}_{human}$. We detail each component of our method below, starting with our pipeline for data collection (Section IV-A), followed by the model architecture design (Section IV-B) and finally with our training strategy (Section IV-C). Our design choices are motivated by the goal of explicitly extracting only the components from simulation and human data that are most suitable for robot learning.

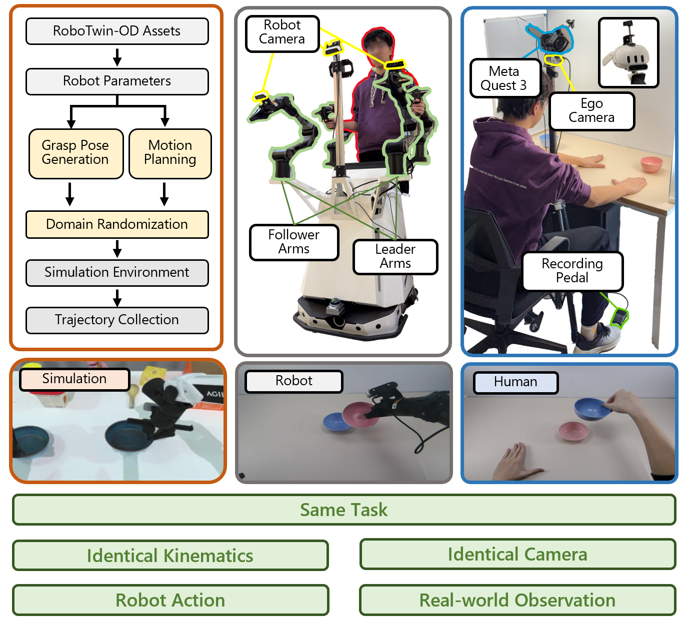

Fig. 2: **Hardware configurations for data collection.** (a) The Real Robot platform (COBOT Magic) employs leader-follower teleoperation with egocentric and wrist views. (b) We design a stationary, cost-effective platform for scalable human data collection. A Meta Quest 3 is mounted on a tripod for fatigue-free 6-DoF pose tracking. A camera hardware-identical to the robot's is adopted to ensure visual isomorphism.

A. Data Collection Pipeline

In this section, we outline a data collection strategy designed to structurally align both simulation and human data with the target real-world domain. We first describe our alignment protocols for human and simulation data collection, and then present the data processing strategy employed to unify these heterogeneous inputs.
Simulation Data Generation. As illustrated in the left panel of Figure 2, we leverage the physics-based platform RoboTwin2.0 [7] to generate large-scale robotic interaction data. To ensure the applicability of this synthetic data, we first employ robot assets that share identical kinematics with the physical hardware. Furthermore, we ensure task consistency by performing manipulation tasks strictly aligned with real-world specifications. This approach yields kinematically feasible trajectories that provide robust robot action priors for our policy learning.
Human Data Collection. As depicted in the right panel of Figure 2, we design a dedicated data collection system to align the observation space of human demonstrations with the robot. To mitigate visual mismatches, we utilize an RGB camera identical to the robot's sensor, mounted on a stationary tripod rather than a wearable device. This setup explicitly removes human ego-motion noise and aligns the recording perspective with the robot's static base. Simultaneously, we

utilize a VR interface (Meta Quest 3) to capture high-precision human hand poses as paired action labels. To streamline the workflow, we incorporate a recording pedal to control the start and stop phases, significantly enhancing collection efficiency. Notably, the entire hardware suite costs under $500, offering a highly accessible and scalable solution for large-scale data acquisition. By integrating these hardware-aligned and cost-effective designs, we establish a scalable pipeline for harvesting diverse human demonstrations that are intrinsically consistent with the robot's observation space.

Action Space Unifying and Data Processing. To effectively fuse complementary priors, we implement a multi-stage alignment strategy for action representations. First, we establish a shared action space by extracting a 14-dimensional end-effector pose vector (position and quaternion) for both sources, using the human wrist as a proxy. These actions are formulated as relative trajectories [25] to decouple them from absolute coordinates. Next, we incorporate embodiment-specific details: appending binary gripper states for robot/simulation data ($a_t^R \in \mathbb{R}^{16}$) and relative fingertip positions for human data ($a_t^H \in \mathbb{R}^{44}$). Finally, we enforce temporal alignment by recording human data at 30Hz while processing it at 10Hz, effectively scaling the execution speed to match the robot's dynamics.

B. Unified Diffusion with Source-dependent Modules

As shown in Figure 3(a), We propose a unified framework that separates domain-specific feature processing from shared policy learning. The architecture is centered around a **shared DiT Backbone**, inspired by [44], which predicts action trajectories conditioned on domain-agnostic feature representations. To enable this unified training, we introduce **Source-dependent Modules** designed to explicitly capture the distinct characteristics of simulation and human data.

For visual inputs, we employ Domain-Specific Visual Adaptors to disentangle the distinct visual information inherent in each domain. Images are first processed by a common Shared Vision Encoder to extract general features, which are then refined by distinct adaptors—a Real-World Adaptor or a Simulation Adaptor—to encode source-specific visual priors (e.g., photorealism vs. synthetic rendering) before entering the backbone.

For action representations, we introduce Embodiment-Specific Action Projectors to separately process the heterogeneous kinematic structures of human hands (44-DoF) and robotic grippers (16-DoF). Specifically, we utilize distinct Encoders to encode human and robot proprioceptive states into the shared latent space, and corresponding Decoders to reconstruct the latent outputs back into valid, embodiment-specific actions. This design allows the backbone to learn a generalized policy distribution by leveraging these distinct yet complementary priors.

C. Sim-and-Human Co-Training recipe

We formalize our learning process into a two-stage pipeline designed to first acquire a generalist manipulation prior and subsequently adapt it to the real robot system.

Hybrid Pre-training via Balanced Sampling. In this stage, we jointly train the unified policy on $\mathcal{D}_{sim}$ and $\mathcal{D}_{human}$ to learn a shared latent representation. To mitigate optimization bias caused by data scale disparities, we set the sampling ratio to $\alpha = \mathbf{0.5}$. This strategy ensures that each training batch consists of approximately 50% human and 50% simulation data. Formally, we train the entire network end-to-end using the standard noise prediction objective. For each optimization step, the total loss is defined as the sum of diffusion losses from both domains:

$$\begin{aligned}\mathcal{L}_{total} = &\mathbb{E}_{(x,a)\sim\mathcal{D}_{sim}}\left[\|\epsilon - \epsilon_\theta(z_t, t, c_{sim})\|^2\right] \\ &+ \mathbb{E}_{(x,a)\sim\mathcal{D}_{human}}\left[\|\epsilon - \epsilon_\theta(z_t, t, c_{human})\|^2\right]\end{aligned} \tag{4}$$

where $\epsilon \sim \mathcal{N}(0, I)$ represents the sampled noise, z_t denotes the noisy latent action at timestep t, and ϵ_θ is the noise predicted by the diffusion backbone conditioned on the domain-specific visual and proprioceptive embeddings (c_{sim} or c_{human}). This strategy enforces a strict equilibrium between learning diverse visual priors from human data and kinematic priors from simulation.

Reality-grounded Module Grafting. Following pre-training, we transition to real-world fine-tuning by restructuring the policy components via a strategic grafting process. As illustrated in Figure 3(b), we construct the target policy by inheriting the *Real-World Vision Adaptor* from the human stream to capture photorealistic visual priors, while simultaneously grafting the *Robot Encoder and Decoder* from the simulation stream to guarantee precise kinematic alignment. Crucially, we explicitly discard the misaligned modules (e.g., human action projectors and simulation visual adaptors). This hybrid architecture is then fine-tuned on the limited dataset $\mathcal{D}_{real}$ with the identical noise prediction objective. By effectively filtering out domain-incompatible components, this strategy seamlessly integrates visual realism with kinematic validity, enabling data-efficient adaptation.

V. Experiments

In this section, we detail the experimental setup designed to validate the proposed framework. We introduce the task suite, evaluation protocols, and the baseline methods used for comparison.

A. Evaluation Tasks

We evaluate our approach on four diverse manipulation tasks, each designed to assess specific aspects of bimanual manipulation: sequential manipulation, fine-grained precision, bimanual coordination, and long-horizon planning. In the *Stack Bowls Two* task, the robot retrieves two bowls from different locations and stacks one atop the other, testing sequential object interaction and alignment. In *Click Bell*, the robot must accurately position the gripper tip to actuate a small button on a service bell, evaluating fine-grained precision control. The *Grab Roller* task involves grasping a long roller stick simultaneously with both grippers to lift it horizontally, demonstrating synchronized bimanual coordination. Finally, in

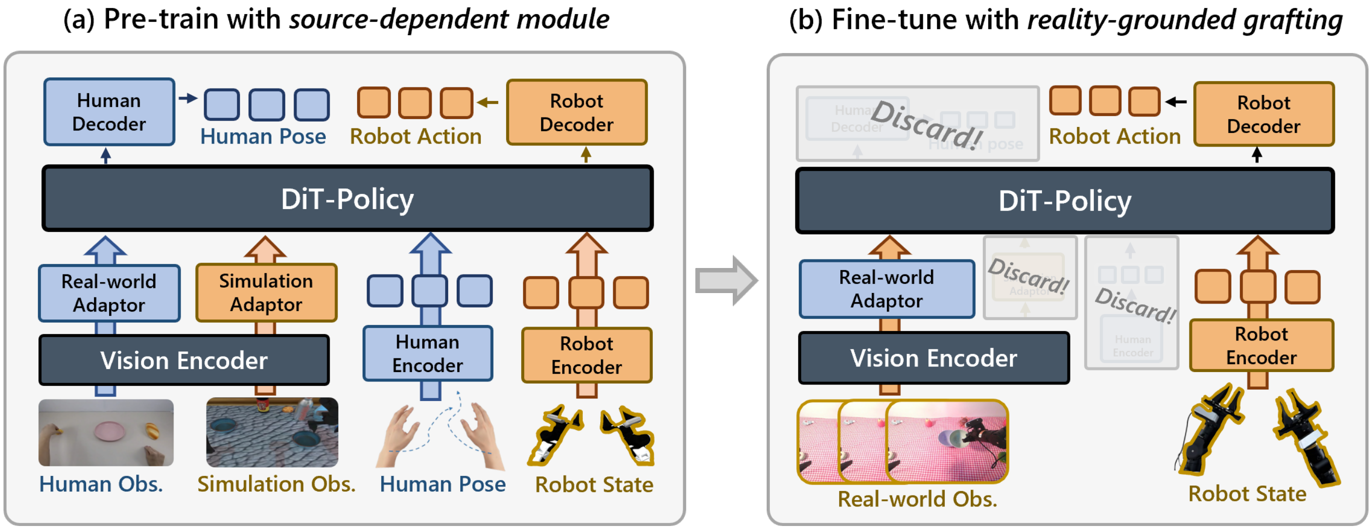

Fig. 3: **Model Overview.** Our approach consists of two main stages: (1) hybrid pre-training that disentangles visual and action representations from simulation and human data to establish a generalized manipulation prior, and (2) real-robot fine-tuning that pairs the human vision adaptor with the robot action encoder and decoder for efficient adaptation.

Put Bread Cabinet, the robot coordinates grasping a piece of bread with one arm while manipulating a drawer handle with the other to perform insertion, assessing composite long-horizon planning. Refer to Appendix VIII-B for additional details about these tasks.

B. Evaluation Protocol

We evaluate the model's performance under two distinct settings. For detailed environmental configurations, experimental protocols, and visual illustrations, please refer to Appendix VIII-D.

- **In-Distribution (ID) Setting.** This setting evaluates the policy within the four specific scenarios (comprising 1 base and 3 complex setups) encountered during the real-robot fine-tuning phase. We conduct 10 evaluation trials for each scenario with randomized initial object poses, resulting in a total of 40 trials.
- **Out-of-Distribution (OOD) Setting.** To assess zero-shot generalization, this setting features two novel "extreme complex" scenarios constructed by recombining environmental factors $\{\mathcal{F}_{obj}, \mathcal{F}_{bg}, \mathcal{F}_{light}, \mathcal{F}_{dist}\}$. Crucially, we impose a strict constraint that all utilized factor instances (e.g., background textures and distractor objects) are *held-out* from any training sources. We perform 10 trials per scenario with randomized initialization, totaling 20 trials.

For both settings, we report the average metrics aggregated across all trials. Specifically, we evaluate the **Success Rate (SR)** to measure final task completion and the **Progress Rate (PR)** to capture partial success in multi-stage tasks, with detailed formulation provided in Appendix VIII-C.

C. Data Collection Protocol

We now detail the data collection protocol designed to systematically construct diverse training scenarios via the recombination of environment factors $\{\mathcal{F}_{obj}, \mathcal{F}_{bg}, \mathcal{F}_{light}, \mathcal{F}_{dist}\}$. For comprehensive details regarding the collection pipeline, object instances, and visual illustrations of the scenarios, please refer to Appendix VIII-E. The data composition for each source is summarized below:

- **Simulation Data.** Utilizing the RoboTwin2.0 [7] framework, we generated 500 kinematically feasible trajectories per task. To enhance visual diversity, we applied the framework's default Domain Randomization techniques to vary textures and lighting.
- **Human Data.** We collected 500 human demonstrations across 12 varied scenarios using a viewpoint-aligned collection pipeline. This diverse dataset allows the model to acquire robust photorealistic visual priors from the real world.
- **Real-world Data.** For fine-tuning, we strictly constrained the dataset to 80 episodes: 50 from a base environment and 30 from three complex scenarios (10 episodes each). All models are fine-tuned using this identical dataset to ensure a fair comparison.

D. Baselines

To validate the effectiveness of our proposed framework, we compare **SimHum** against three baseline variants. For a fair comparison, all methods utilize the identical backbone architecture and are optimized using the same hyperparameters. Please refer to Appendix VIII-H and Appendix VIII-I for detailed specifications regarding the network architecture and the training hyperparameters. The baselines differ only in their data sources and training strategies:

- **Real-Only:** A policy initialized randomly and trained *from scratch* solely on the real-robot dataset. This serves as a lower bound to assess the necessity of pre-training.
- **SimR:** A policy pre-trained on simulation data followed by fine-tuning on real-world data. This variant isolates

the contribution of simulation priors and tests the direct sim-to-real transferability.

- **HumR:** A policy pre-trained on human demonstrations. Due to the action space mismatch, the state encoder and action decoder are re-initialized before fine-tuning on real data. This variant isolates the contribution of human priors and evaluates the efficacy of learning from human demonstrations.
- **SimHum (Ours):** Our full method is jointly pre-trained on both simulation and human datasets, followed by fine-tuning on real-robot data. This paradigm effectively unifies diverse data sources to enable robust real-world generalization.

VI. Results and Analysis

In this section, we empirically validate the proposed sim-and-human co-training framework. Our experiments are designed to answer the following four key research questions:

> *(1) Does co-training on simulation and human data enhance unseen generalization?*
> *(2) What specific knowledge does the policy transfer from simulation and human data, respectively?*
> *(3) Does the method outperform baselines in sample efficiency and scaling behavior?*
> *(4) What architectural decisions and data mixing strategies yield the optimal performance?*

A. Effectiveness of Sim-and-Human Co-Training

To rigorously evaluate the effectiveness of SimHum, we benchmark it against three baselines defined in Section V-D: Real-Only, HumR, and SimR. Following the protocol in Section V-B, we assess all models under both ID and OOD settings to quantify task-solving capabilities and zero-shot generalization, respectively.

As reported in Table I, our method achieves superior performance by synthesizing the distinct advantages of each data source to construct a comprehensive manipulation prior. **Firstly**, regarding source complementarity, SimHum significantly outperforms single-source baselines (SimR and HumR). By explicitly extracting kinematic precision from simulation and visual realism from human data, SimHum overcomes the inherent limitations of isolated domains (e.g., the visual gap in SimR or the embodiment gap in HumR). This synergy allows the policy to leverage the best of both worlds, yielding the highest success rates in both ID (82.1% vs. 42.5%/69.2%) and OOD (62.5% vs. 22.5%/27.5%) scenarios. **Secondly**, regarding real-world generalization, SimHum demonstrates remarkable robustness compared to the Real-Only baseline. While Real-Only achieves reasonable ID performance (40.0%), it suffers from severe overfitting in unseen environments, collapsing to 8.8%. In stark contrast, SimHum maintains a 62.5% success rate—surpassing Real-Only by a factor of **7.1**×. This indicates that the unified knowledge base constructed from heterogeneous data prevents overfitting to spurious features, enabling the policy to learn invariant representations essential for robust zero-shot adaptation.

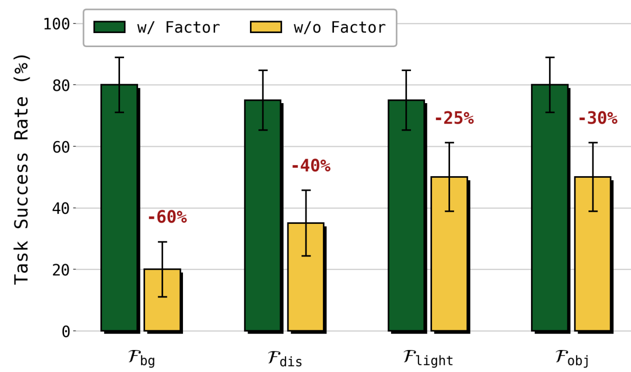

Fig. 4: **Human data diversity enables robust environmental generalization.** We compare the average Success Rate on Stack Bowls Two between SimHum trained on the full human dataset (w/ Factor) and subsets excluding specific environmental factors (w/o Factor). Models are evaluated in OOD scenarios corresponding to the omitted factors (see Figure 15).

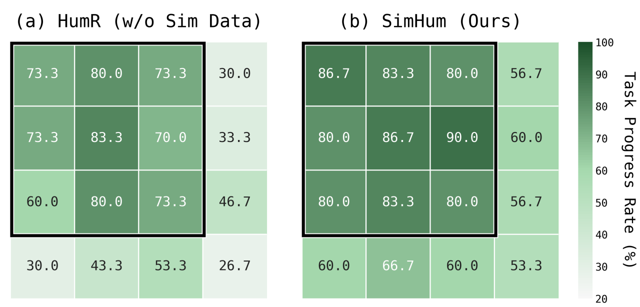

Fig. 5: **Simulation data enhances the generalization to unseen positions.** We visualize the position generalization performance by comparing the Human-only baseline (HumR) and our method (SimHum) on the Stack Bowls Two task. Both policies are fine-tuned using real-robot data collected strictly within a restricted 3×3 zone (indicated by the black box), while evaluations are conducted across an extended 4×4 grid.

B. Respective Contributions of Simulation and Human Data

To identify the specific knowledge transferred from simulation and human data, we systematically decouple the contributions of these data source through factor-wise ablation and position generalization analyses on the *Stack_Bowls_Two*.

Human data is critical for environmental generalization. To quantify the impact of human demonstrations on environmental robustness, we conduct a factor-wise ablation study on the factors defined in Section III-B. As detailed in Appendix VIII-F, we train SimHum with human data subsets that systematically exclude specific environmental factors and evaluate the policy's generalization performance in the respective OOD scenarios compared to the full-data baseline. The results in Figure 4 reveal that removing factors invariably degrades performance. Most notably, ablating *Background* and *Distractor* causes precipitous drops of **60**%

TABLE I: **Effectiveness of Pre-train on both simulation and human data.** We compare the performance under In-Distribution and Out-of-Distribution settings, with specific environmental configurations detailed in Figure V-B. The results report the mean Success Rate (SR) and Task Progress Rate (TPR) averaged over evaluation trials.

Method	Stack Bowls Two		Click Bell		Put Bread Cabinet		Grab Roller		Average	
	SR ↑	PR ↑	SR ↑	PR ↑	SR ↑	PR ↑	SR ↑	PR ↑	SR ↑	PR ↑
In-Distribution Setting										
Real-Only	53 ± 8	74 ± 7	48 ± 8	65 ± 8	28 ± 7	48 ± 8	33 ± 7	50 ± 8	40.0	59.4
HumR	58 ± 8	80 ± 6	55 ± 8	63 ± 8	30 ± 7	52 ± 8	28 ± 7	53 ± 8	42.5	61.7
SimR	65 ± 8	87 ± 5	55 ± 8	74 ± 7	35 ± 8	55 ± 8	35 ± 8	61 ± 8	47.5	69.2
Ours	**78 ± 7**	**90 ± 5**	**70 ± 7**	**84 ± 6**	**65 ± 8**	**78 ± 7**	**58 ± 8**	**76 ± 7**	**67.5**	**82.1**
Out-of-Distribution Setting										
Real-Only	15 ± 8	48 ± 11	15 ± 8	35 ± 11	5 ± 5	23 ± 10	0	20 ± 9	8.8	31.7
HumR	30 ± 10	58 ± 11	25 ± 10	43 ± 11	20 ± 9	45 ± 11	15 ± 8	28 ± 10	22.5	43.3
SimR	40 ± 11	60 ± 11	40 ± 11	53 ± 11	20 ± 9	42 ± 11	10 ± 7	40 ± 11	27.5	48.5
Ours	**75 ± 10**	**83 ± 8**	**60 ± 11**	**80 ± 9**	**55 ± 11**	**68 ± 10**	**60 ± 11**	**78 ± 9**	**62.5**	**77.3**

and 40% respectively, confirming the necessity of human data for visual complexity. In conclusion, the realistic visual priors inherent in human data are indispensable for achieving robust environmental generalization.

Simulation data enhances policy robustness to novel positions. To investigate the value of simulation data, we evaluate generalization across the spatial initialization factor $\mathcal{F}_{init}$. As detailed in Appendix VIII-G, we adopt an extrapolation protocol where policies are fine-tuned on real-robot data restricted to a central 3×3 zone, while testing is conducted on a larger 4×4 grid that includes unseen outer positions. The results indicate that the *HumR* policy (Figure 16a) suffers from rapid performance decay in unseen regions, suggesting that while human data provides visual realism, it lacks the robot-specific kinematic knowledge needed for spatial extrapolation. In contrast, *SimHum* (Figure 16b) demonstrates consistent improvements in both seen and unseen zones. We attribute this robustness to the dense kinematic priors from simulation, which enable the policy to predict feasible and IK-solvable poses, even for spatial regions not covered during training.

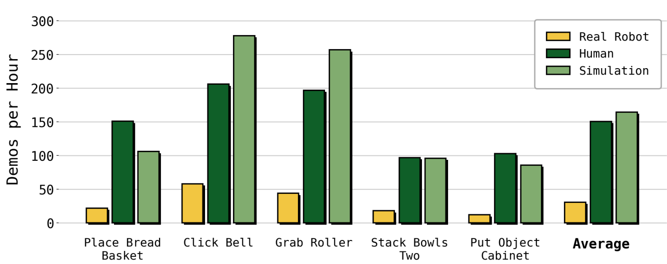

Fig. 6: **Human and simulation pipelines are 5× faster than robot teleoperation.** We compare the collection speeds of Real Robot, Human, and Simulation pipelines across five tasks. Note that simulation speed is evaluated using a single-threaded process on an NVIDIA RTX 4090 GPU.

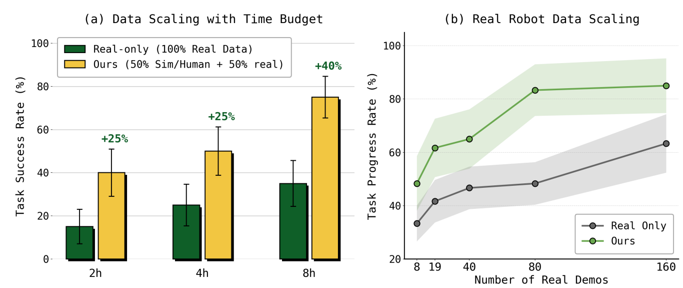

Fig. 7: **(a) Data Scaling with Time Budget.** We evaluate policies under fixed total data collection time limits (2h, 4h, and 8h). The "Real-only" baseline utilizes 100% of the budget for real robot data, while our method allocates 50% to simulation/human data and 50% to real data. **(b) Real Robot Data Scaling.** We assess performance by varying the number of real robot demonstrations from 8 to 160. Shaded areas indicate the standard deviation across multiple trials.

C. Data Efficiency and Scaling

We evaluate the generalization and scaling capabilities of our framework under OOD settings. All experiments are conducted on the **Stack Bowls Two** task, where we report the performance metrics averaged across multiple trials.

Simulation and human data collection are significantly faster than robot teleoperation. We compare the data collection speed across different sources. As shown in Figure 6, collecting Human or Simulation data is approximately **5× faster** than Real Robot teleoperation (averaged across tasks), where the simulation data is generated by a single NVIDIA RTX 4090 GPU. Crucially, human data acquisition is robot-agnostic, offering significant potential for parallel scaling across multiple human operators. Furthermore, as simulation data generation is fully automated, it can proceed in parallel with human data collection, maximizing total throughput.

TABLE II: Ablation study of different components. We evaluate the contribution of each design module on the Stack Bowls Two task under OOD settings. Metrics reported include Success Rate (SR) and Task Progress Rate (PR).

Method	SR ↑	PR ↑
SimHum w/o vision adaptor	40 ± 15	68 ± 15
SimHum w/o real-world adaptor	55 ± 16	70 ± 14
SimHum w/o relative action	45 ± 16	58 ± 16
SimHum (Ours)	$\mathbf{75 \pm 14}$	$\mathbf{83 \pm 12}$

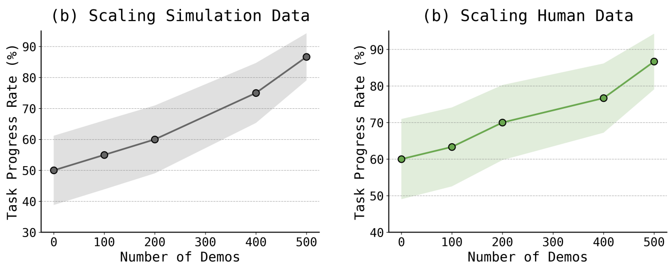

Fig. 8: **Synergistic Scaling of Simulation and Human Data.** We systematically vary the volume of one data source (ranging from 0 to 500) while maintaining the other source fixed at 500. All experiments are conducted under OOD settings on the Stack Bowls Two task, with shaded regions denoting the standard deviation across independent evaluation trials.

SimHum achieves superior data efficiency under time constraints. We compare the performance of our method against a real-only baseline in OOD settings under fixed total data collection time budgets (2h, 4h, and 8h).Specifically, our strategy allocates 50% of the time budget to collect human and simulation data for policy pre-training, followed by utilizing the remaining 50% to collect real robot data for fine-tuning. In contrast, the baseline utilizes 100% of the budget exclusively for real robot data collection. As illustrated in Fig. 7(a), our SimHum yields significantly better results within the same timeframe. Notably, as the total time budget extends to 8 hours, our method achieves a +40% higher success rate compared to the baseline. This confirms that leveraging heterogeneous data for pre-training is a far more efficient strategy for improving policy performance than simply extending the duration of real-robot data collection.

Superior real-world data efficiency. We further evaluate the sample efficiency of our approach in OOD settings by varying the number of real robot demonstrations from 8 to 160. To capture finer-grained trends in policy execution, we report the task progress rate as shown in Fig. 7(b). The results indicate that our method maintains a substantial lead over the Real-only baseline across all data regimes. Notably, the performance gain peaks at 80 demonstrations, where our approach achieves a +35% improvement over the baseline. This demonstrates that under identical data constraints, the representations learned from our heterogeneous pre-training enable the policy to achieve significantly higher completion quality than training from scratch with real data alone.

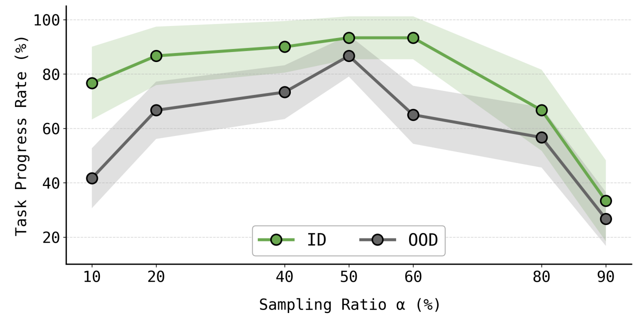

Fig. 9: **Balanced Sampling Yields Optimal Generalization.** We evaluate the average Task Progress Rate on the Stack Bowls Two task under ID and OOD settings. The sampling ratio α represents the probability of sampling a human demonstration during batch construction, where shaded regions denote standard deviation.

D. Ablation Study and Co-Training Recipe

In this section, we systematically validate our architectural design choices and investigate the optimal data co-training strategy. All ablation studies are conducted under the OOD setting on the Stack Bowls Two task, unless otherwise explicitly stated. For quantitative evaluation, we report the average task Progress Rate (PR) aggregated over multiple independent trials to measure the policy's performance.

Impact of Architectural Components. We quantify the contribution of each design component in Table II. **First**, removing the vision adaptor precipitates the most severe performance collapse (SR $-\mathbf{35}\%$, PR $-\mathbf{15}\%$), indicating that without explicit feature disentanglement, the distributional shift between simulation and real-world data creates destructive interference. **Second**, conducting fine-tuning with the simulation adaptor (i.e., w/o real-world adaptor) diminishes performance (SR $-\mathbf{20}\%$, PR $-\mathbf{13}\%$). This underscores the critical necessity of the real-world vision adaptor for successfully transferring photorealistic visual priors to the real robot. **Finally**, ablating the relative action formulation leads to a substantial decline (SR $-\mathbf{30}\%$, PR $\mathbf{25}\%$), validating its essential role in unifying the distinct absolute coordinate systems of heterogeneous data.

Sim and Human data are complementary in the recipe. We investigate the synergy between the two data sources by systematically scaling the volume of one source while holding the other constant (Figure 8). We observe that increasing either Human or Simulation data volume independently yields consistent performance gains. This phenomenon validates that Human and Simulation data contribute complementary knowledge, confirming that both sources are indispensable for constructing the optimal training recipe.

Optimal Batch Sampling Strategy. We identify the optimal batch sampling strategy by analyzing the impact of the sampling ratio αm, which controls the percentage of human data in a batch. As shown in Figure 9, the performance trends reveal the distinct contributions of each data source.**First,**

when simulation data is marginalized, performance degrades in both ID and OOD settings. This indicates that simulation data provides the necessary kinematic priors for basic robot control. **Second,** when human data is reduced, a significant decline is only observed in OOD settings. This suggests that human data contributes critical visual priors that are essential for visual generalization to novel environments. Consequently, our analysis reveals that $\alpha = 0.5$ yields the optimal equilibrium, effectively synergizing these complementary priors.

VII. Conclusion and Limitations

In this work, we introduced SimHum, a unified co-training framework that synthesizes complementary knowledge from heterogeneous sources to achieve data-efficient and generalizable robotic manipulation. By explicitly integrating the robot kinematics prior from simulation with the photorealistic visual prior from human data, SimHum constructs a comprehensive policy prior that overcomes the limitations of isolated data domains. Empirical results demonstrate the efficacy of this synergy: our method surpasses the real-only baseline by 40% under a fixed 8-hour budget while achieving a 62.5% success rate in OOD scenarios—exceeding the baseline by a factor of $7.1\times$. Furthermore, we offer a principled co-training recipe, providing actionable insights for scaling heterogeneous robot learning.

Despite these advancements, four limitations define our future roadmap. First, our current evaluation primarily focuses on basic manipulation primitives such as pick-and-place. Future work should extend to more challenging scenarios, including dexterous manipulation and long-horizon tasks. Second, while SimHum significantly improves robustness, the performance gap in OOD settings suggests that further investigation is needed to achieve near-perfect reliability. Thirdly, our current model is a visuomotor policy with no multi-tasking capabilities. Integrating our heterogeneous recipe with Vision-Language-Action (VLA) models would allow us to investigate their potential for open-world instruction following and cross-task transfer. Finally, we currently rely on curated datasets. Scaling up pre-training with diverse in-the-wild data represents a promising avenue to explore emergent behaviors in robotic foundation models.

Acknowledgments

References

[1] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell *et al.*, "Language models are few-shot learners," in *Advances in Neural Information Processing Systems*, 2020, pp. 1877–1901. 1

[2] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat *et al.*, "GPT-4 technical report," *arXiv preprint arXiv:2303.08774*, 2023. 1

[3] A. Yang, B. Yu, C. Li, D. Liu, F. Huang, H. Huang, J. Jiang, J. Tu, J. Zhang, J. Zhou *et al.*, "Qwen2.5-1m technical report," *arXiv preprint arXiv:2501.15383*, 2025. 1

[4] DeepSeek-AI, A. Liu, B. Feng, B. Xue, B. Wang, B. Wu, C. Lu, C. Zhao *et al.*, "DeepSeek-V3 technical report," *arXiv preprint arXiv:2412.19437*, 2024. 1

[5] A. Mandlekar, S. Nasiriany, B. Wen, I. Akinola, Y. Narang, L. Fan, Y. Zhu, and D. Fox, "MimicGen: A data generation system for scalable robot learning using human demonstrations," in *Annual Conference on Robot Learning*, 2023. 1, 2

[6] Y. Mu, T. Chen, Z. Chen, S. Peng, Z. Lan, Z. Gao, Z. Liang, Q. Yu, Y. Zou, M. Xu *et al.*, "RoboTwin: Dual-arm robot benchmark with generative digital twins," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 27 649–27 660. 1, 2

[7] T. Chen, Z. Chen, B. Chen, Z. Cai, Y. Liu, Q. Liang, Z. Li, X. Lin, Y. Ge, Z. Gu *et al.*, "Robotwin 2.0: A scalable data generator and benchmark with strong domain randomization for robust bimanual robotic manipulation," *arXiv preprint arXiv:2506.18088*, 2025. 1, 2, 3, 5, 14

[8] Z. Jiang, Y. Xie, K. Lin, Z. Xu, W. Wan, A. Mandlekar, L. J. Fan, and Y. Zhu, "DexMimicGen: Automated data generation for bimanual dexterous manipulation via imitation learning," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 16 923–16 930. 1

[9] R.-Z. Qiu, S. Yang, X. Cheng, C. Chawla, J. Li, T. He, G. Yan, D. J. Yoon, R. Hoque, L. Paulsen *et al.*, "Humanoid policy ˜ human policy," *Conference on Robot Learning (CoRL)*, 2025. 1

[10] R. Yang, Q. Yu, Y. Wu, R. Yan, B. Li, A.-C. Cheng, X. Zou, Y. Fang, X. Cheng, R.-Z. Qiu *et al.*, "EgoVLA: Learning vision-language-action models from egocentric human videos," *arXiv preprint arXiv:2507.12440*, 2025. 1, 2

[11] T. Tao, M. K. Srirama, J. J. Liu, K. Shaw, and D. Pathak, "DexWild: Dexterous human interactions for in-the-wild robot policies," 2025. 1, 2

[12] S. Kareer, D. Patel, R. Punamiya, P. Mathur, S. Cheng, C. Wang, J. Hoffman, and D. Xu, "EgoMimic: Scaling imitation learning via egocentric video," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 13 226–13 233. 1, 2

[13] D. A. Pomerleau, "Alvinn: an autonomous land vehicle in a neural network," in *Advances in Neural Information Processing Systems*, 1988. 2

[14] T. Osa, J. Pajarinen, G. Neumann, J. A. Bagnell, P. Abbeel, and J. Peters, "An algorithmic perspective on imitation learning," *Found. Trends Robotics*, pp. 1–179, 2018. 2

[15] B. D. Argall, S. Chernova, M. Veloso, and B. Browning, "A survey of robot learning from demonstration," *Robotics and Autonomous Systems*, pp. 469–483, 2009. 2

[16] Octo Model Team, D. Ghosh, H. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari, J. Hejna, C. Xu, J. Luo, T. Kreiman, Y. Tan, P. Sanketi, Q. Vuong, T. Xiao, D. Sadigh, C. Finn, and S. Levine, "Octo: An open-source generalist robot policy," in *Proceedings of Robotics: Science and Systems*, 2024. 2

[17] S. Liu, L. Wu, B. Li, H. Tan, H. Chen, Z. Wang, K. Xu, H. Su, and J. Zhu, "RDT-1b: a diffusion foundation model for bimanual manipulation," 2025. 2

[18] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, "Diffusion Policy: Visuomotor Policy Learning via Action Diffusion," in *Robotics: Science and Systems (RSS)*, 2023. 2

[19] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware," in *Robotics: Science and Systems (RSS)*, 2023. 2

[20] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, J. Dabis, C. Finn, K. Gopalakrishnan, K. Hausman, A. Herzog, J. Hsu *et al.*, "Rt-1: Robotics transformer for real-world control at scale," in *Robotics: Science and Systems (RSS)*, 2023. 2

[21] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choromanski, T. Ding, D. D. Arenas, A. Dubey, C. Finn *et al.*, "Rt-2: Vision-language-action models transfer web knowledge to robotic control," in *Conference on Robot Learning (CoRL)*, 2023. 2

[22] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. P. Foster, P. R. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn, "OpenVLA: An open-source vision-language-action model," in *Conference on Robot Learning (CoRL)*, 2024. 2

[23] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter *et al.*, "π_0: A Vision-Language-Action flow model for general robot control," *arXiv preprint arXiv:2410.24164*, 2024. 2

[24] K. Black, N. Brown, J. Darpinian, K. Dhabalia, D. Driess, A. Esmail, M. R. Equi, C. Finn, N. Fusai, M. Y. Galliker, D. Ghosh, L. Groom, K. Hausman, brian ichter, S. Jakubczak, T. Jones, L. Ke, D. LeBlanc, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, A. Z. Ren, L. X. Shi, L. Smith, J. T. Springenberg, K. Stachowicz, J. Tanner, Q. Vuong, H. Walke, A. Walling, H. Wang, L. Yu, and U. Zhilinsky, "$\pi_{0.5}$: a Vision-Language-Action model with Open-World generalization," in *Annual Conference on Robot Learning*, 2025. 2

[25] C. Chi, Z. Xu, C. Pan, E. Cousineau, B. Burchfiel, S. Feng, R. Tedrake, and S. Song, "Universal manipulation interface: In-the-wild robot teaching without in-the-wild robots," in *Robotics: Science and Systems (RSS)*, 2024. 2, 4

[26] Zhaxizhuoma, K. Liu, C. Guan, Z. Jia, Z. Wu, X. Liu, T. Wang, S. Liang, P. CHEN, P. Zhang, H. Song, D. Qu, D. Wang, Z. Wang, N. Cao, Y. Ding, B. Zhao, and X. Li, "FastUMI: A scalable and hardware-independent universal manipulation interface with dataset," 2025. 2

[27] X. Cheng, J. Li, S. Yang, G. Yang, and X. Wang, "Open-television: Teleoperation with immersive active visual feedback," in *Conference on Robot Learning*, 2024. 2

[28] Y. Ze, Z. Chen, J. P. Araujo, Z. ang Cao, X. B. Peng, J. Wu, and K. Liu, "TWIST: Teleoperated whole-body imitation system," in *Conference on Robot Learning*, 2025. 2

[29] C. Wang, H. Shi, W. Wang, R. Zhang, L. Fei-Fei, and K. Liu, "DexCap: Scalable and portable mocap data collection system for dexterous manipulation," in *Robotics: Science and Systems (RSS)*, 2024. 2

[30] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," in *2017 IEEE/RSJ international conference on intelligent robots and systems (IROS)*, 2017, pp. 23–30. 2

[31] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, "Sim-to-real transfer of robotic control with dynamics randomization," in *2018 IEEE international conference on robotics and automation (ICRA)*, 2018, pp. 3803–3810. 2

[32] NVIDIA, J. Bjorck, F. Castañeda, N. Cherniadev, X. Da, R. Ding, L. Fan, Y. Fang, D. Fox, F. Hu *et al.*, "GR00T N1: an open foundation model for generalist humanoid robots," *CoRR*, 2025. 2

[33] S. Nasiriany, A. Maddukuri, L. Zhang, A. Parikh, A. Lo, A. Joshi, A. Mandlekar, and Y. Zhu, "Robocasa: Large-scale simulation of everyday tasks for generalist robots," in *Robotics: Science and Systems (RSS)*, 2024. 2

[34] K. Grauman, A. Westbury, E. Byrne, Z. Chavis, A. Furnari, R. Giraldo, M. Hamburger, H. Hao, J. Pan, R. Piramuthu *et al.*, "Ego4D: Around the world in 3,000 hours of egocentric video," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. 2

[35] S. Nair, A. Rajeswaran, V. Kumar, C. Finn, and A. Gupta, "R3m: A universal visual representation for robot manipulation," in *Conference on Robot Learning (CoRL)*, 2022. 2

[36] Y. J. Ma, S. Sodhani, D. Jayaraman, O. Bastani, V. Kumar, and A. Zhang, "VIP: Towards universal visual reward and representation via value-implicit pre-training," in *International Conference on Learning Representations (ICLR)*, 2023. 2

[37] A. Padalkar, A. Pooley, A. Jain, A. Bewley, A. Herzog, A. Irpan, A. Khazatsky, A. Rai, A. Singh, A. Brohan *et al.*, "Open x-embodiment: Robotic learning datasets and rt-x models," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024. 2

[38] L. Zha, A. Badithela, M. Zhang, J. Lidard, J. Bao, E. Zhou, D. Snyder, A. Z. Ren, D. Shah, and A. Majumdar, "Guiding data collection via factored scaling curves," in *Second Workshop on Out-of-Distribution Generaliza-*

tion in Robotics at RSS 2025, 2025. 2

[39] W. Pumacay, I. Singh, J. Duan, R. Krishna, J. Thomason, and D. Fox, "The COLOSSEUM: A benchmark for evaluating generalization for robotic manipulation," in *RSS 2024 Workshop: Data Generation for Robotics*, 2024. 2

[40] A. Xie, L. Lee, T. Xiao, and C. Finn, "Decomposing the generalization gap in imitation learning for visual robotic manipulation," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 3153–3160. 2

[41] J. Gao, A. Xie, T. Xiao, C. Finn, and D. Sadigh, "Efficient data collection for robotic manipulation via compositional generalization," in *RSS 2024 Workshop: Data Generation for Robotics*, 2024. 2

[42] J. Hejna, C. Bhateja, Y. Jian, K. Pertsch, and D. Sadigh, "Re-mix: Optimizing data mixtures for large scale imitation learning," *CoRR*, 2024. 2

[43] V. Saxena, M. Bronars, N. R. Arachchige, K. Wang, W. C. Shin, S. Nasiriany, A. Mandlekar, and D. Xu, "What matters in learning from large-scale datasets for robot manipulation," in *The Thirteenth International Conference on Learning Representations*, 2025. 2

[44] S. Dasari, O. Mees, S. Zhao, M. K. Srirama, and S. Levine, "The ingredients for robotic diffusion transformers," *arXiv preprint arXiv:2410.10088*, 2024. 4, 15

VIII. Appendix

A. Overview

The Appendix is organized as follows:

- **Task Definitions and Scoring Criteria** (Appendix VIII-B): Details the definitions of the four manipulation tasks and their milestone-based scoring protocols.
- **Evaluation Metrics** (Appendix VIII-C): Describes the quantitative metrics (Success Rate and Task Progress Rate) used for policy assessment.
- **Evaluation Settings** (Appendix VIII-D): Specifies the In-Distribution (ID) and Out-of-Distribution (OOD) settings for evaluation.
- **Data Collection Protocol** (Appendix VIII-E): Elaborates on the data collection process, object instances, and environmental variations across Simulation, Human, and Real-Robot data.
- **Protocol for Human Data Factor Ablation** (Appendix VIII-F): Details the experimental strategy to analyze how human data diversity contributes to robustness across different environmental factors.
- **Protocol for Position Generalization Analysis** (Appendix VIII-G): Outlines the evaluation protocol designed to assess the specific impact of simulation data on the policy's spatial generalization capabilities.
- **Implementation Details** (Appendix VIII-H): Provides specific architectural details of the Source-dependent Modules, visual encoders, and the diffusion backbone.
- **Training Hyperparameters** (Appendix VIII-I): Lists the exact optimization settings and training schedules for pre-training and fine-tuning.

B. Task Definitions and Scoring Criteria

We evaluate our proposed **SimHum** framework on four distinct manipulation tasks: Stack Bowls Two, Put Bread Cabinet, Click Bell, and Grab Roller. As visualized in Fig. 10, these tasks are carefully selected to cover a wide range of motor skills, including bimanual coordination, long-horizon planning, precision actuation, and dynamic stability.

To rigorously quantify the policy's performance beyond simple binary success, we designed a milestone-based scoring protocol for each task. This discretized scoring system allows us to distinguish between policies that fail completely and those that achieve partial success (e.g., successful grasping but failed manipulation), providing deeper insights into the specific capabilities learned by the model.

Stack Bowls Two (Max Score: 3). This task requires precise dual-arm coordination to stack two bowls initially placed apart. The scoring reflects the cumulative success of independent arm actions and their coordination.

- **0:** Failure to grasp any bowl or meaningful interaction.
- **1:** Successfully grasping the left bowl with the left arm.
- **2:** Successfully grasping both bowls (one in each hand).
- **3:** Precisely stacking one bowl into the other without dropping either.

Put Bread Cabinet (Max Score: 3). This is a long-horizon task involving articulated object manipulation and multi-stage planning.

- **0:** Failure to grasp the bread object from the table.
- **1:** Successfully grasping the bread object.
- **2:** Successfully pull the cabinet door open.
- **3:** Placing the bread inside the cabinet and releasing it.

Click Bell (Max Score: 2). This task assesses the agent's fine motor control and precision in actuating a small target.

- **0:** Failure to reach the target area.
- **1:** Moving the end-effector to directly above the bell.
- **2:** Successfully applying downward force to depress the button and produce a ringing sound.

Grab Roller (Max Score: 2). This task evaluates bimanual synchronization and grasp stability on cylindrical objects.

- **0:** Failure to establish a stable grasp on the roller.
- **1:** Successfully establishing a grasp on the roller stick with both grippers.
- **2:** Lifting the roller to the target height with both arms while maintaining grasp without the object falling.

C. Evaluation Metrics

To measure both the final task completion and the quality of partial execution, we employ the following metrics. The specific milestone definitions and maximum scores (S_{max}) for each task are detailed in Appendix VIII-B.

1) Success Rate (SR): SR measures the percentage of trials where the policy achieves the maximum possible score (i.e., complete task success). It is calculated as:

$$SR = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(S_i = S_{max}) \times 100\% \tag{5}$$

where N is the total number of evaluation trials, S_i represents the score achieved in the i-th trial, S_{max} is the maximum score defined for the task, and $\mathbb{I}(\cdot)$ is the indicator function.

2) Task Progress Rate (PR): PR provides a fine-grained evaluation of the policy's ability to progress through multi-stage tasks, distinguishing between complete failures and partial successes. It is defined as the ratio of the total accumulated score across all trials to the maximum possible total score:

$$PR = \frac{\sum_{i=1}^{N} S_i}{N \times S_{max}} \times 100\% \tag{6}$$

A higher PR indicates that the policy consistently completes more stages of the task, even if it fails to achieve final success.

D. Evaluation Settings

We classify the evaluation scenarios into In-Distribution (ID) and Out-of-Distribution (OOD) settings based on the environmental factors defined in Section III-B. Visual examples of these settings are provided in Fig. 11. For each evaluation trial, we randomly place the objects within the spatial region specified in Fig. 14, which is consistent with the area used for data collection.

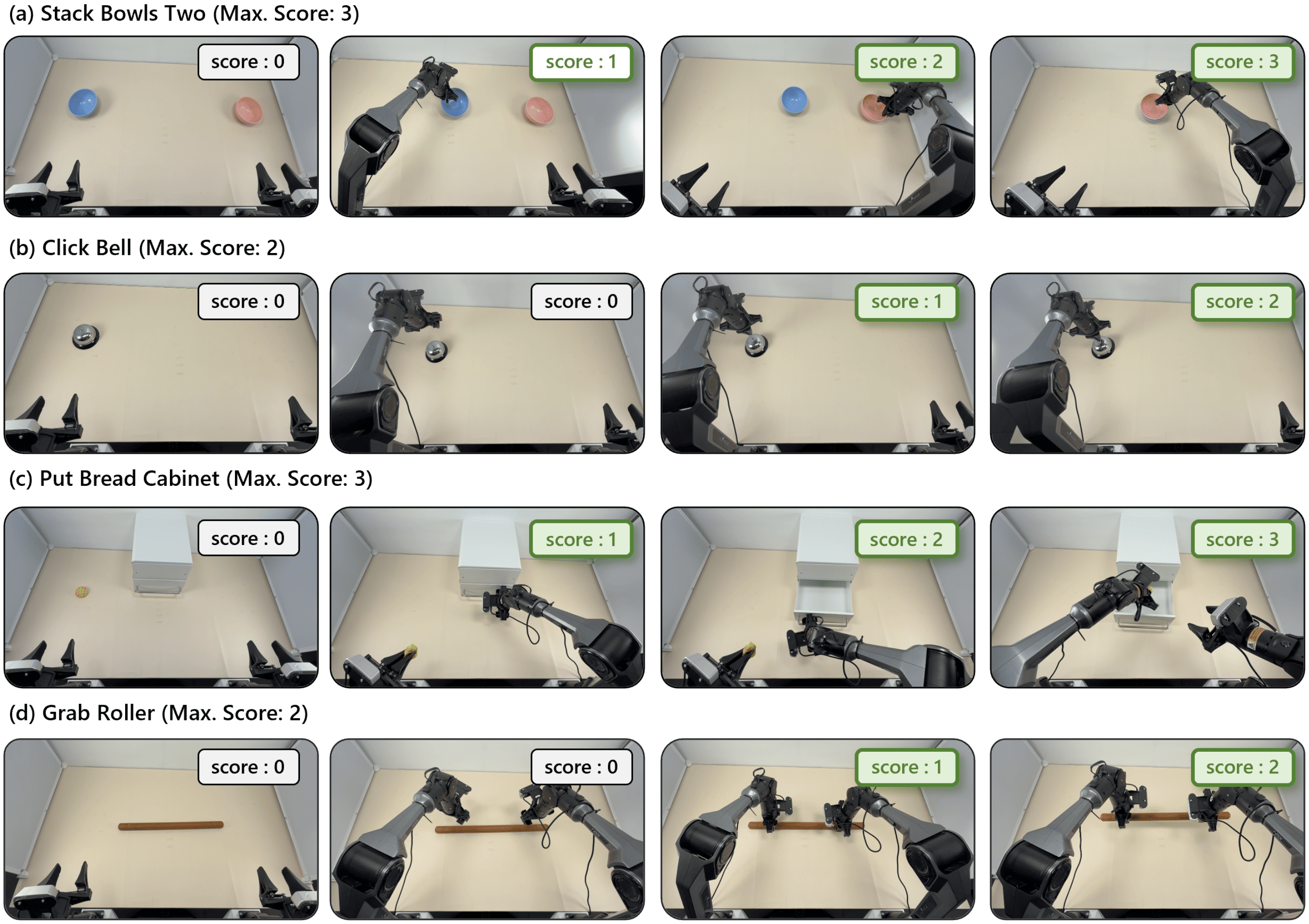

Fig. 10: **Task Definitions and Scoring.** Visual breakdown of the four manipulation tasks (Stack Bowls, Click Bell, Put Bread, Grab Roller) and their respective success/progress criteria.

1) In-Distribution (ID): This setting evaluates the policy's performance in environments seen during the real-robot fine-tuning phase. As illustrated in Figure 11(a), it consists of two categories of scenarios:

- **Base Scenario (1 scenario):** A standard environment characterized by a clean white background and stable, neutral lighting conditions, serving as the baseline for robot operation.
- **Complex Scenarios (3 scenarios):** These scenarios introduce domain variations encountered during training. They feature backgrounds with diverse textures, a moderate number of distractor objects, and dynamic interference such as flashing blue lights.

Regarding the objects, the target items manipulated in the ID setting correspond strictly to the object instances seen during training, as depicted in the left panel of Fig. 11(c). Crucially, all four ID scenarios appeared in the robot fine-tuning dataset. For each task, we conduct 10 evaluation trials per scenario, resulting in a total of 40 trials. We report the aggregated average SR and PR across these 40 trials.

2) Out-of-Distribution (OOD): To test zero-shot generalization, we construct two novel scenarios labeled as "Extreme Complex" (Fig. 11(b)). In contrast to the ID settings, these scenarios are constructed entirely from strictly **held-out** environmental factors that never appeared in either the Human or Real-robot datasets. Specifically, we introduce novel background textures (featuring irregular wrinkles to increase complexity), unseen distractor objects arranged in higher density to create heavy clutter, and held-out task object instances (as highlighted in Fig. 11(c)). We perform 10 evaluation trials for each of the two OOD scenarios, totaling 20 trials, and report the average SR and PR.

E. Data Collection Protocol

In this section, we elaborate on the data collection process across Simulation, Human, and Real-Robot domains. The collection pipeline is guided by the environmental factors defined in Section III-B. Specifically, the object instances ($\mathcal{F}_{obj}$) utilized for each domain are visualized in Fig. 13, while the valid spatial initialization ranges ($\mathcal{F}_{init}$) for all tasks are illustrated in Fig. 14. To construct diverse environments,

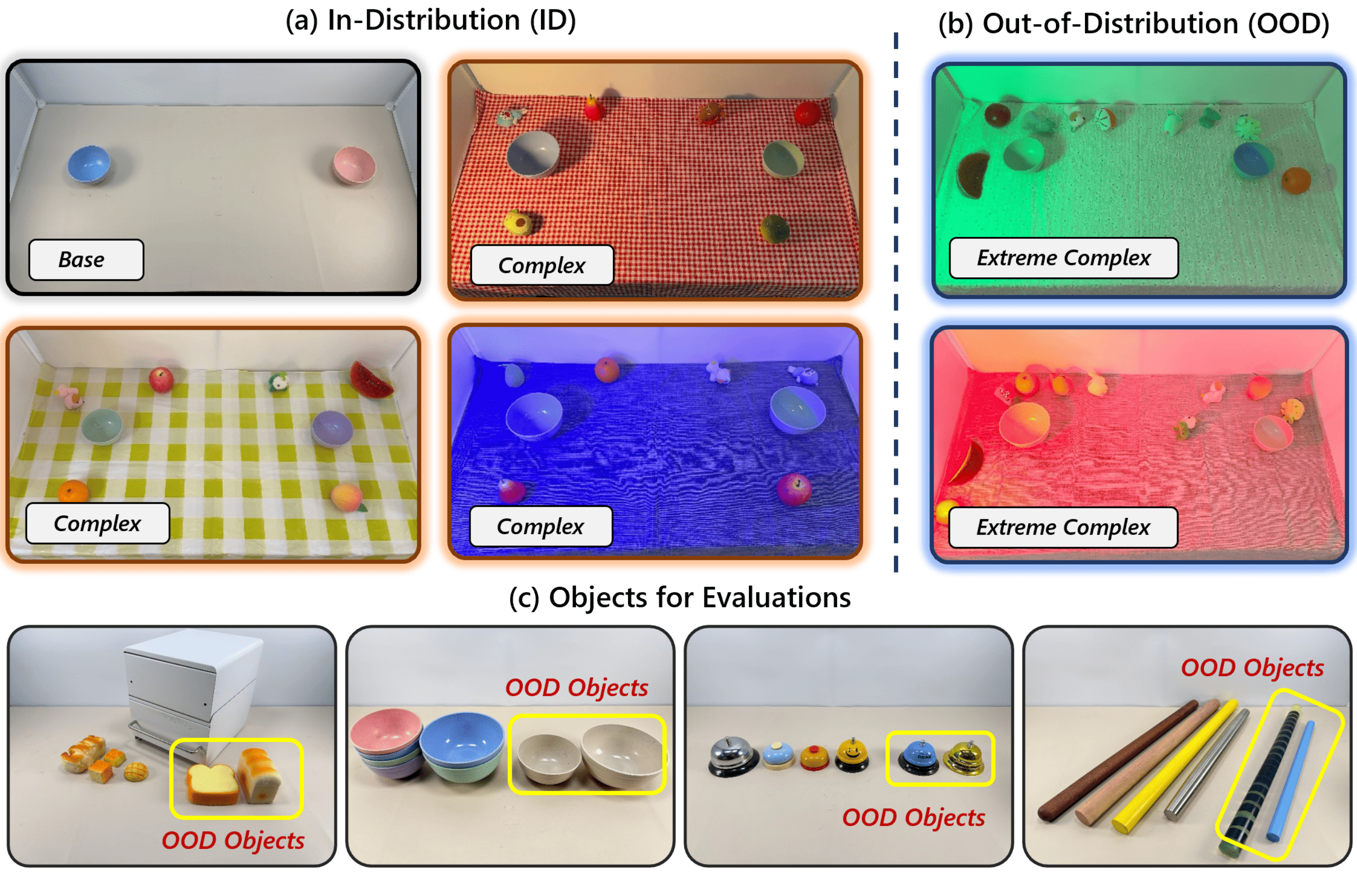

Fig. 11: **Evaluation Settings.** (a) **In-Distribution (ID)** scenarios comprising a standard *Base* environment and *Complex* environments seen during training. (b) **Out-of-Distribution (OOD)** scenarios labeled as *Extreme Complex*, constructed using strictly held-out environmental factors (e.g., unseen lighting, novel textures). (c) **Evaluation Objects** across tasks, where yellow bounding boxes indicate specific held-out instances (OOD Objects) introduced to test zero-shot generalization.

we generate distinct scenarios by combinatorially varying visual distractors ($\mathcal{F}_{dist}$), illumination ($\mathcal{F}_{light}$), and background ($\mathcal{F}_{bg}$). The specific configurations for these heterogeneous scenarios are detailed as follows:

For **Simulation**, we utilize RoboTwin 2.0 [7] and employ its built-in Domain Randomization engine to randomize textures, lighting, and distractors, yielding the sampled observations shown in Fig. 12(a). For **Human data**, to capture diverse real-world visual distributions, we curated 12 distinct scenarios by permuting environmental factors (Fig. 12(b)); these setups encompass combinations of 4 tablecloth textures, 6 lighting conditions, and 16 distinct visual distractors to introduce realistic occlusion and clutter. Finally, for **Real-Robot data**, to emulate a realistic data-scarce fine-tuning regime given the prohibitive cost of data collection, we strategically limited the design to 4 representative scenarios (comprising 1 base and 3 complex setups). As visualized in Fig. 12(c), these setups utilize combinations of 3 tablecloths, 3 lighting conditions, and 12 visual distractors to evaluate transfer capabilities.

F. Protocol for Human Data Factor Ablation

In this section, we detail the experimental protocol used to quantify the specific contributions of human data diversity toward environmental robustness, as discussed in Section VI-B. To rigorously isolate the impact of each environmental factor ($\mathcal{F}_{bg}$, $\mathcal{F}_{dist}$, $\mathcal{F}_{light}$, $\mathcal{F}_{obj}$), we designed a "Leave-One-Factor-Out" ablation strategy. The core logic of this experiment is illustrated in Fig. 15. We systematically construct training subsets by filtering the full human dataset to exclude diversity associated with a specific target factor, thereby forcing the model to rely on a limited "canonical" distribution for that dimension.

For a target factor (e.g., Background $\mathcal{F}_{bg}$), we first remove all human demonstrations that exhibit variations in that factor (e.g., excluding all trajectories collected on diverse tablecloths). The resulting training subset contains only the "base" condition for that specific factor (e.g., the default table surface) while retaining diversity in all other factors. Subsequently, we evaluate the policy trained on this restricted subset in unseen OOD scenarios that specifically feature the omitted factor. For instance, in the $\mathcal{F}_{bg}$ ablation, the model is evaluated on surfaces with novel textures (tablecloths) that were absent

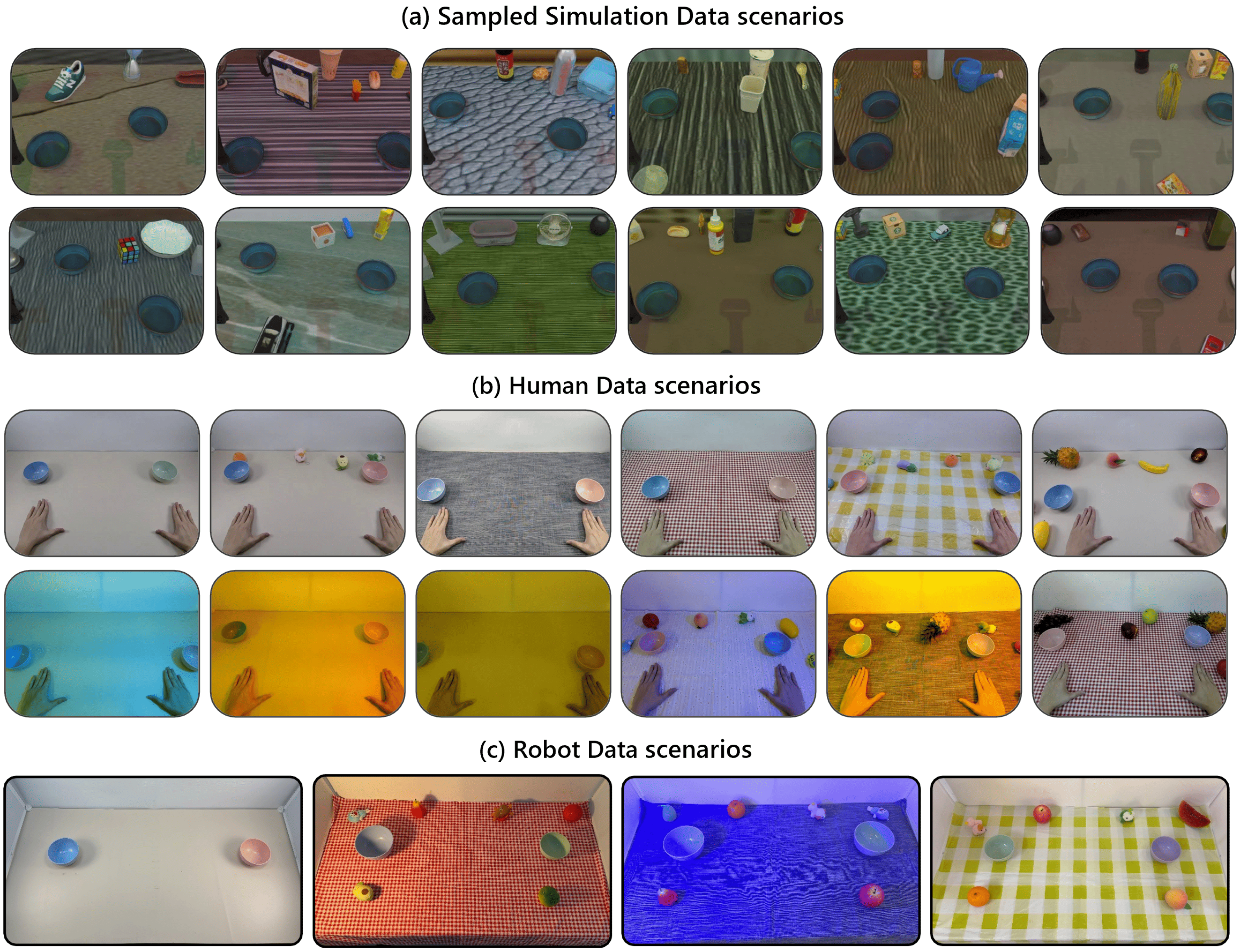

Fig. 12: **Visualization of Data Collection Scenarios.** These diverse environments are constructed by systematically varying visual distractors ($\mathcal{F}_{dist}$), illumination ($\mathcal{F}_{light}$), and background ($\mathcal{F}_{bg}$). (a) Simulation environments leveraging extensive Domain Randomization to synthesize visual diversity. (b) Diverse human demonstration setups constructed by varying these environmental factors. (c) Real-robot setups ranging from simple, clean base scenarios to complex environments with significant clutter and lighting variations.

from its training distribution. This protocol ensures that any performance degradation compared to the full-data baseline can be directly attributed to the lack of visual priors for that specific environmental factor.

G. Protocol for Position Generalization Analysis

In this section, we outline the experimental protocol used to evaluate the impact of simulation data on spatial robustness ($\mathcal{F}_{init}$). We utilize the Stack Bowls Two task, defining a discretized 4×4 spatial grid on both left and right workspaces to precisely control object placement. Real robot data is strictly collected within the inner 3×3 zones (Green) using the visual scenarios depicted in Fig. 12(c). Evaluation is conducted across the full 4×4 grid, effectively probing the policy's performance on unseen outer coordinates (Orange). We conduct 10 independent evaluation trials for each of the 16 spatial positions and report the average Task Progress Rate.

H. Implementation Details

We implemented our framework using the PyTorch library and managed training configurations via Hydra. All experiments, including simulation data generation and policy training, were conducted on a workstation equipped with a single NVIDIA RTX 4090 GPU.

Visual Encoder Architecture. To process visual observations, we employed a ResNet-18 backbone initialized with ImageNet-1K pre-trained weights. To capture distinct spatial features, we employ independent visual encoders for each camera view without weight sharing. The extracted feature maps are flattened, concatenated, and projected into the transformer's embedding dimension.

Diffusion Policy Architecture. Our policy architecture employs a shared DiT Backbone consistent with [44]. The diffusion process follows the Denoising Diffusion Probabilistic

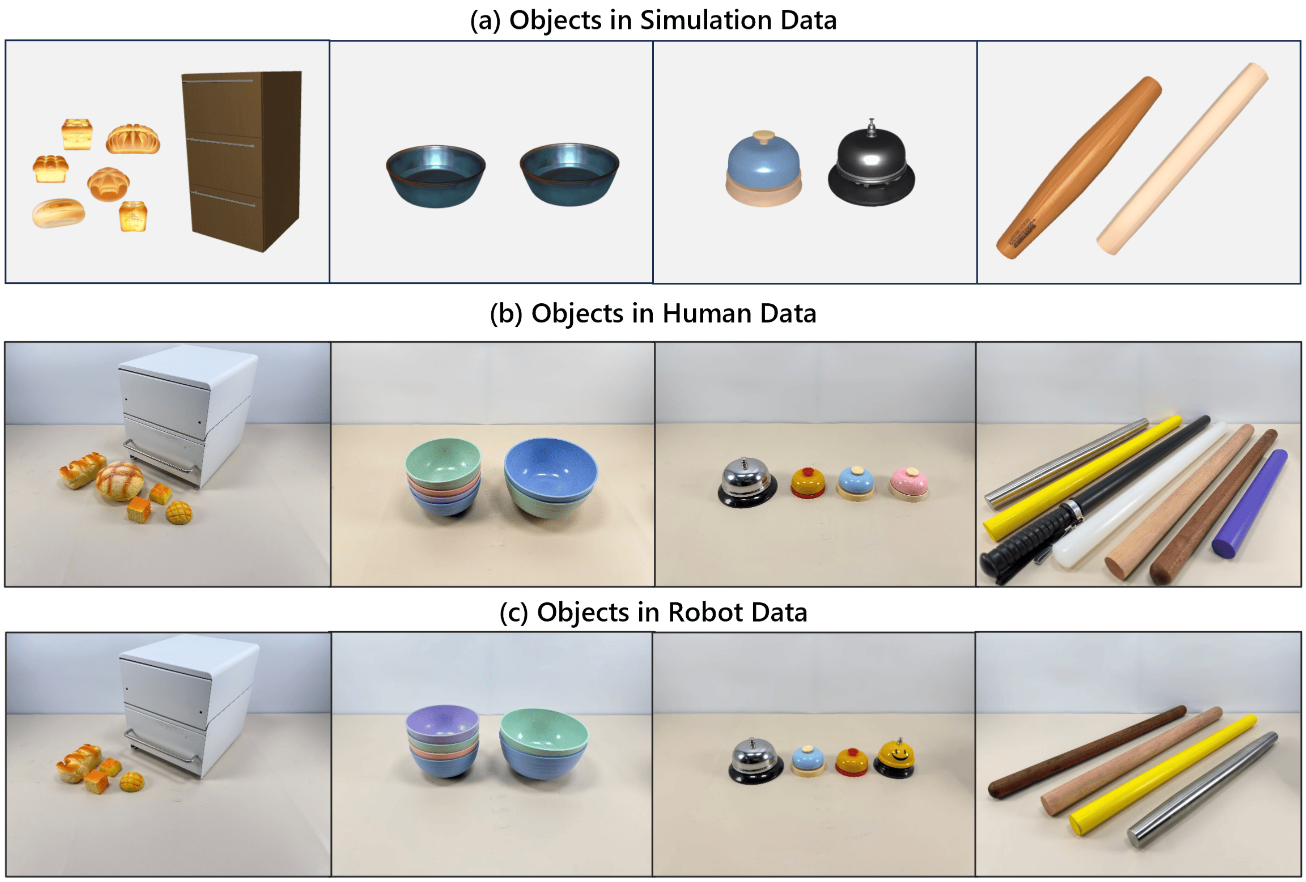

Fig. 13: **Objects used in data collection.** Visualization of the specific object instances ($\mathcal{F}_{obj}$) utilized across different tasks. (a) Synthetic assets from the simulation dataset. (b)Real-world objects from the human dataset. (c) Real-world objects from the real-robot dataset.

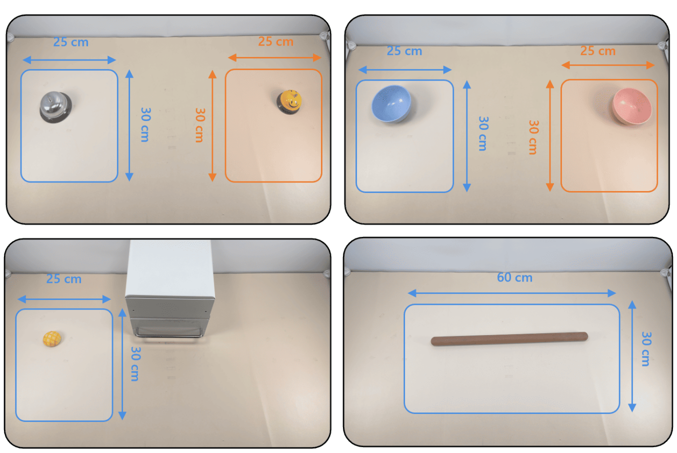

Fig. 14: **Object Initialization Ranges.** Illustration of the spatial coverage ranges used for randomizing object placement ($\mathcal{F}_{init}$) across the four manipulation tasks. The bounding boxes indicate the valid sampling zones (with dimensions in cm) where objects are initialized during both data collection and evaluation trials.

Models (DDPM), utilizing a squared cosine beta schedule with 100 diffusion steps during training, which is accelerated to 8 steps during inference. To enable unified training on heterogeneous data, we introduce Source-dependent Modules designed to capture the distinct characteristics of each domain. The implementation details of these modules are as follows:

- **Domain-Specific Visual Adaptors:** To disentangle the distinct visual information between the real world and simulation, we employ separate vision adaptors. Both adaptors are two-layer MLPs with GELU activations that project high-dimensional visual features into the latent space. Specifically, we utilize a Real-World Adaptor to process the embeddings from human ego-centric videos. In contrast, for the simulation domain, we instantiate independent Simulation Adaptors for each camera view to handle the multi-view synthetic data.
- **Embodiment-Specific Encoders:** We utilize separate projection layers to align heterogeneous kinematic spaces before they enter the shared backbone. Proprioceptive State Encoders project the robot state (16-dim) and human state (44-dim) to the hidden dimension using a linear layer preceded by Dropout ($p = 0.2$) to mitigate

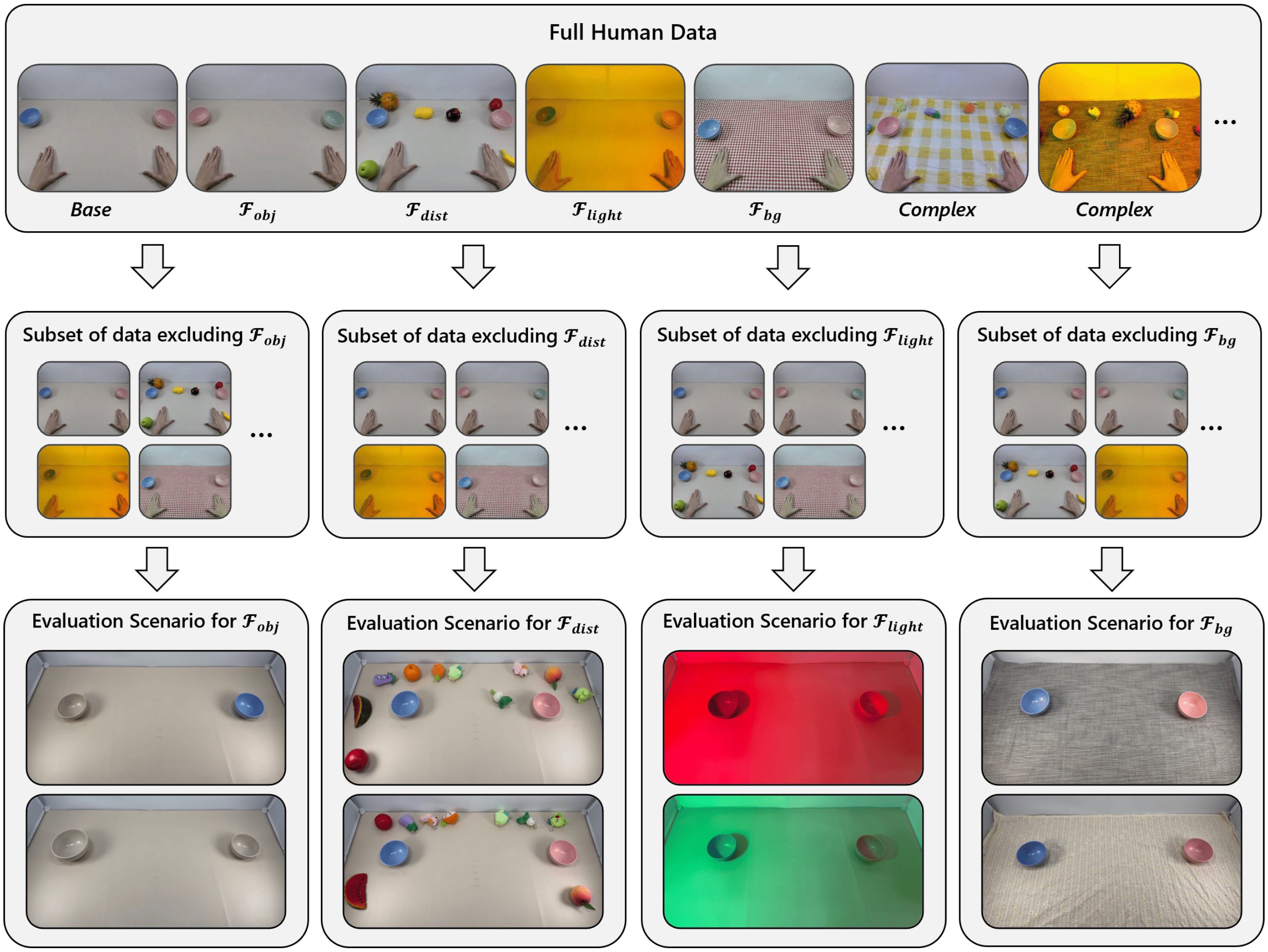

Fig. 15: **Workflow for Environmental Factor Ablation.** We systematically isolate the contribution of specific factors by curating training subsets (Middle Row) that exclude data variations corresponding to a target factor (e.g., removing all tablecloth data for $\mathcal{F}_{bg}$). We then evaluate the model in targeted OOD scenarios (Bottom Row) that explicitly require robustness to the omitted factor, comparing performance against the full human dataset (Top Row).

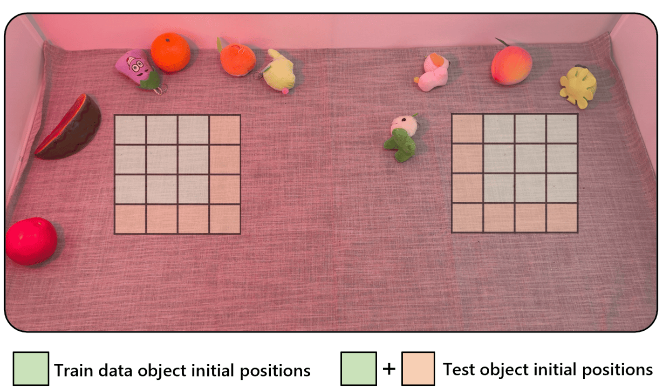

Fig. 16: **Position Generalization Protocol.** Illustration of the discretized 4×4 evaluation grids used in the Stack Bowls Two task. Real-robot training data is strictly confined to the inner 3×3 zones (Green), while evaluation tests the policy's ability to extrapolate to the unseen outer positions (Orange).

overfitting. Action Projectors encode the noisy action inputs during diffusion using a non-linear MLP (Linear $\rightarrow$ GELU $\rightarrow$ Linear) to effectively map the distinct action manifolds of human hands and robotic grippers into the shared latent space.

- **Domain-Aware Output Heads:** The latent embeddings from the final backbone layer are projected into action space via separate heads. Specifically, distinct linear layers transform the shared features into 16-dimensional robot action noise and 44-dimensional human action noise, respectively.

I. Training Hyperparameters

In this section, we outline the specific hyperparameters used for the optimization process. Table III summarizes the settings for both the pre-training and fine-tuning stages, including the optimizer configuration, learning rate schedule, and the number of training iterations.

TABLE III: Optimization Hyperparameters

Hyperparameter	**Pre-training**	**Fine-tuning**
Training Steps	200,000	60,000
Optimizer	AdamW	AdamW
Learning Rate	1×10^{-4}	5×10^{-5}
Optimizer Betas	(0.95, 0.999)	(0.95, 0.999)
Weight Decay	1×10^{-6}	1×10^{-6}
LR Scheduler	Cosine Decay	Cosine Decay
Warmup Steps	2000	2000